Calculus II Project

Objective: Find the general solution of the differential equation $y'' - 4y' + 13y = 0$.

Example 1

Verify that the general solution $y = C \cdot e^{-4x}$ satisfies the differential equation $y' + 4y = 0$.

Verifying that the general solution $y = C \cdot e^{-4x}$ satisfies the differential equation $y' + 4y = 0$:

$y = C \cdot e^{-4x}$, where C is a constant.

$y' = C \cdot e^{-4x} \cdot (-4) = -4Ce^{-4x}$

Substituting $y(x) = Ce^{-4x}$ and $y'(x) = -4Ce^{-4x}$ into the differential equation $y' + 4y = 0$:

$-4Ce^{-4x} + 4\left[Ce^{-4x} \right] = 0$

$-4Ce^{-4x} + 4Ce^{-4x} = 0$

$0 = 0$

Therefore, $y(x) = Ce^{-4x}$ is the general solution of the differential equation $y' + 4y = 0$.

Example 2

Determine whether the general solution $y = 4x^3 + x + C$ satisfies the differential equation $\frac{1}{2}y'' + \frac{1}{x} \cdot y' = 24x + \frac{1}{x}$.

Note: $y = 4x^3 + x + C$, where C is a constant.

$y' = 12x^2 + 1$

$y'' = 24x$

Substituting $y(x) = 4x^3 + x + C$, $y'(x) = 12x^2 + 1$, and $y''(x) = 24x$ into the differential equation $\frac{1}{2}y'' + \frac{1}{x}y' = 24x + \frac{1}{x}$:

$$\frac{1}{2}(24x) + \frac{1}{x} \cdot (12x^2 + 1) = 24x + \frac{1}{x}.$$

$$12x + 12x + \frac{1}{x} = 24x + \frac{1}{x}.$$

$$24x + \frac{1}{x} = 24x + \frac{1}{x}.$$

Hence, the general solution $y = 4x^3 + x + C$ satisfies the differential equation $\frac{1}{2}y'' + \frac{1}{x} \cdot y' = 24x + \frac{1}{x}$.

Project Part 1:

Note: Answer must be completely legible and clear.
Unclear answer is considered incorrect.

a) Taylor series for $f(x) = e^x$ centered at $x = 0$: $e^x =$ _____?_____

b) Taylor series for $f(x) = \cos x$ centered at $x = 0$: $\cos x =$ _____?_____

c) Taylor series for $f(x) = \sin x$ centered at $x = 0$: $\sin x =$ _____?_____

d) Taylor series for $f(x) = e^{i \cdot x}$ centered at $x = 0$: $e^{i \cdot x} =$ _____?_____, where $i = \sqrt{-1}$.

e) Using part (d), $e^{\beta \cdot i \cdot x} =$? , where $i = \sqrt{-1}$ and β is a real constant.

f) Using part (d), $e^{-\beta \cdot i \cdot x} =$? , where $i = \sqrt{-1}$ and β is a real constant.

g) $e^{\beta \cdot i \cdot x} + e^{-\beta \cdot i \cdot x} =$?

h) $e^{\beta \cdot i \cdot x} - e^{-\beta \cdot i \cdot x} =$?

i) $e^{(\alpha+\beta \cdot i) \cdot x} + e^{(\alpha-\beta \cdot i) \cdot x} = e^{\alpha \cdot x} \cdot e^{\beta \cdot i \cdot x} + e^{\alpha \cdot x} \cdot e^{-\beta \cdot i \cdot x} = e^{\alpha \cdot x} \cdot \left(e^{\beta \cdot i \cdot x} + e^{-\beta \cdot i \cdot x} \right) =$?

Note: Answer has "$\cos(\beta x)$".

j) $e^{(\alpha+\beta\cdot i)\cdot x} - e^{(\alpha-\beta\cdot i)\cdot x} = e^{\alpha\cdot x}\cdot e^{\beta\cdot i\cdot x} - e^{\alpha\cdot x}\cdot e^{-\beta\cdot i\cdot x} = e^{\alpha\cdot x}\cdot\left(e^{\beta\cdot i\cdot x} - e^{-\beta\cdot i\cdot x}\right) =$ _____?_____

Note: Answer has "$\sin(\beta x)$".

Project Part 2:

Note: Answer must be completely legible and clear.
Unclear answer is considered incorrect.

The characteristic or indicial equation for the second-order differential equation

$y'' - 4y' + 13y = 0$ is $1m^2 - 4m + 13 = 0$.

a) The roots of the indicial equation $1m^2 - 4m + 13 = 0$ are $m_1 =$ ___?___ and $m_2 =$ ___?___.

b) m_1 and m_2 should be complex numbers of the form $\alpha \pm \beta \cdot i$.
$\alpha =$ ___?___ and $\beta =$ ___?___

c) Verify that $y_1(x) = e^{(\alpha+\beta\cdot i)\cdot x}$ is a solution of the second-order differential equation $y'' - 4y' + 13y = 0$.

$y_1'(x) =$ ___?___

$y_1''(x) =$ ___?___

$y_1''(x) - 4\left[y_1'(x)\right] + 13\left[y_1(x)\right] =$ ___?___

d) Verify that $y_2(x) = e^{(\alpha-\beta\cdot i)\cdot x}$ is a solution of the second-order differential equation $y'' - 4y' + 13y = 0$.

$y_2'(x) =$?

$y_2''(x) =$?

$y_2''(x) - 4\left[y_2'(x)\right] + 13\left[y_2(x)\right] =$?

e) Verify that $y_3(x) = 2e^{\alpha\cdot x}\cdot\cos(\beta x)$ is a solution of the second-order differential equation $y'' - 4y' + 13y = 0$.

$y_3'(x) =$?

$y_3''(x) =$?

$y_3''(x) - 4\left[y_3'(x)\right] + 13\left[y_3(x)\right] =$?

f) Verify that $y_4(x) = 2e^{\alpha\cdot x}\cdot\sin(\beta x)$ is a solution of the second-order differential equation $y'' - 4y' + 13y = 0$.

$y_4'(x) =$?

$y_4''(x) =$?

$y_4''(x) - 4\left[y_4'(x)\right] + 13\left[y_4(x)\right] =$?

Project Part 3:

Note: Answer must be completely legible and clear.
Unclear answer is considered incorrect.

a) What is the Wronskian determinant?

b) Wronskian determinant of the functions $y_3(x) = 2e^{\alpha x} \cdot \cos(\beta x)$ and $y_4(x) = 2e^{\alpha x} \cdot \sin(\beta x)$ is

$$\begin{vmatrix} e^{\alpha x}\cos(\beta x) & e^{\alpha x}\sin(\beta x) \\ e^{\alpha x}\left[-\beta\sin(\beta x)\right]+\alpha e^{\alpha x}\cos(\beta x) & e^{\alpha x}\left[\beta\cos(\beta x)\right]+\alpha e^{\alpha x}\sin(\beta x) \end{vmatrix} = \underline{\quad ? \quad}$$

c) Use part (b) to explain why the functions $y_3(x) = 2e^{\alpha x} \cdot \cos(\beta x)$ and $y_4(x) = 2e^{\alpha x} \cdot \sin(\beta x)$ are linearly independent.

Project Part 4 (Summary):

Note: Answer must be completely legible and clear.
Unclear answer is considered incorrect.

a) Is the function $y_3(x) = 2e^{\alpha \cdot x} \cdot \cos(\beta x)$ a solution of the second-order differential equation $y'' - 4y' + 13y = 0$?

b) Is the function $y_3(x) = 2e^{\alpha \cdot x} \cdot \sin(\beta x)$ a solution of the second-order differential equation $y'' - 4y' + 13y = 0$?

c) Are the functions $y_3(x) = 2e^{\alpha \cdot x} \cdot \cos(\beta x)$ and $y_4(x) = 2e^{\alpha \cdot x} \cdot \sin(\beta x)$ are linearly independent?

d) The general solution of the second-order differential equation $y'' - 4y' + 13y = 0$ is $y(x) = C_1 \cdot y_3(x) + C_1 \cdot y_4(x) =$ ___?___, where C_1 and C_2 are arbitrary constants.